

MedCare Network

Hospital Operations & Revenue Analysis

Multi-Branch Healthcare Dataset - Exploratory Data Analysis

Prepared for: Healthcare Analytics Portfolio | July 2026

5 linked tables: doctors, patients, appointments, treatments, billing

Introduction & Objective

- MedCare Network operates 3 branches: Central Hospital, Eastside Clinic, Westside Clinic
- 3 specializations covered: Pediatrics, Dermatology, Oncology
- 200 cleaned appointment records linked across doctors, patients, treatments, and billing
- Goal: understand no-show behaviour, revenue leakage, and doctor staffing versus demand
- Tools used: Python, Polars, Pandas, Numpy, Matplotlib, Seaborn, Plotly

Dataset Overview

Metric	Raw Dataset	Cleaned Dataset
Total Rows (all tables)	670	660
Total Tables	5	5
Total Nulls	109	0
Duplicate Rows Found	10	0
Orphaned Foreign Keys	Present (e.g. D999)	Nullled, not guessed

Data Cleaning: Key Decisions

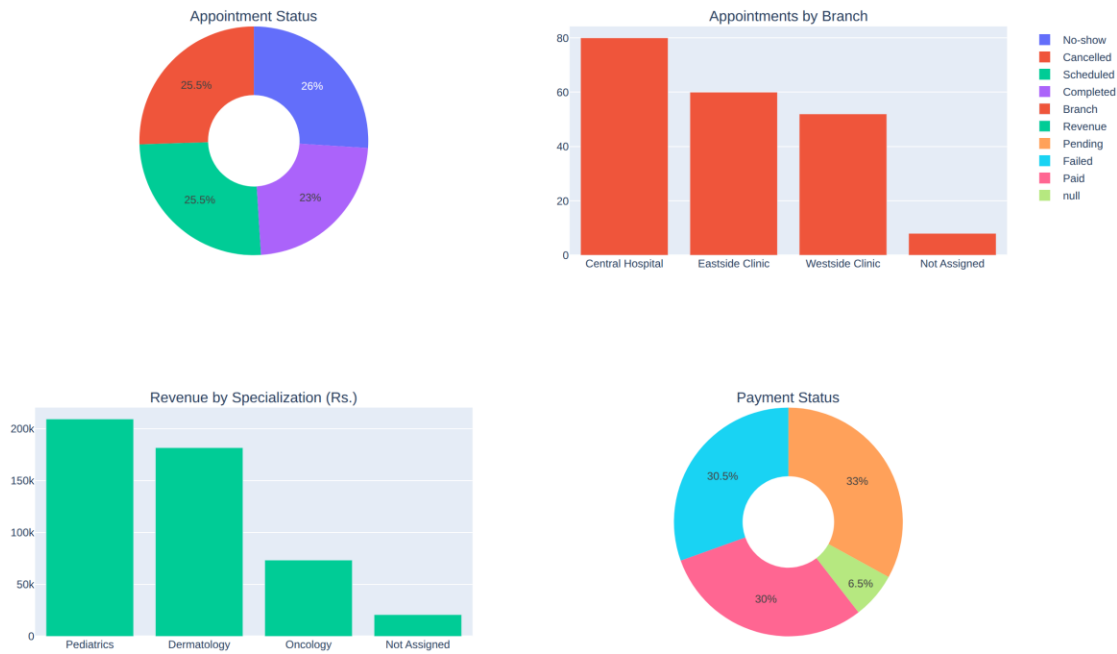
- 10 exact duplicate rows found and removed across all 5 tables
- Text casing standardized (e.g. dermatology vs DERMATOLOGY)
- Dates parsed across 5 mixed formats; invalid ages (110+) nulled, not guessed
- Costs cleaned of currency symbols and commas, negative values corrected with abs()
- 8 missing treatment costs filled using median per treatment type, not dropped
- Invalid doctor IDs (like D999) nulled rather than kept as broken links

Key Metrics Summary

Appointments Analyzed	Overall No-show Rate	Avg Treatment Cost	Total Billed Revenue
200	26.00%	Rs. 2,738.91	Rs. 4,85,224.31

Hospital Overview Dashboard

Hospital Overview Dashboard — Operations (top) & Finance (bottom)



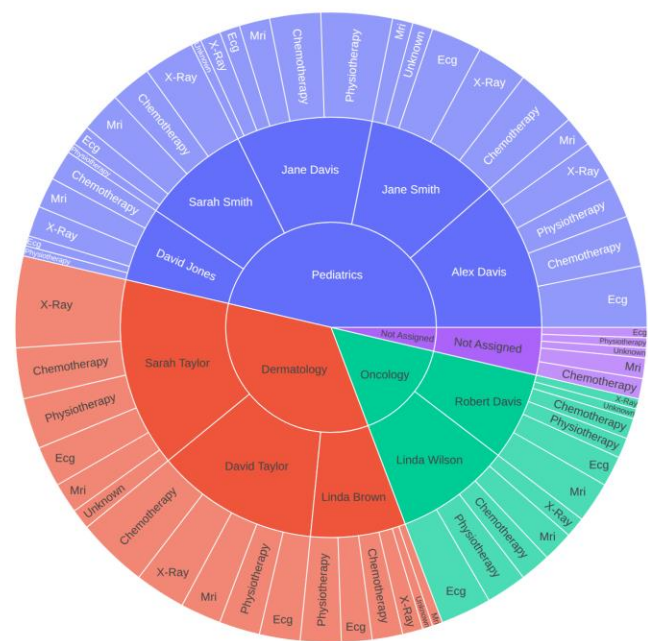
Appointment status, appointments by branch, revenue by specialization, payment status

Dashboard: Key Takeaways

- No-shows are the single largest outcome at 26%, more than completed visits (23%)
- Central Hospital carries the heaviest load: 80 of 200 appointments (40%)
- Pediatrics leads on revenue, but Pending is the largest payment status bucket

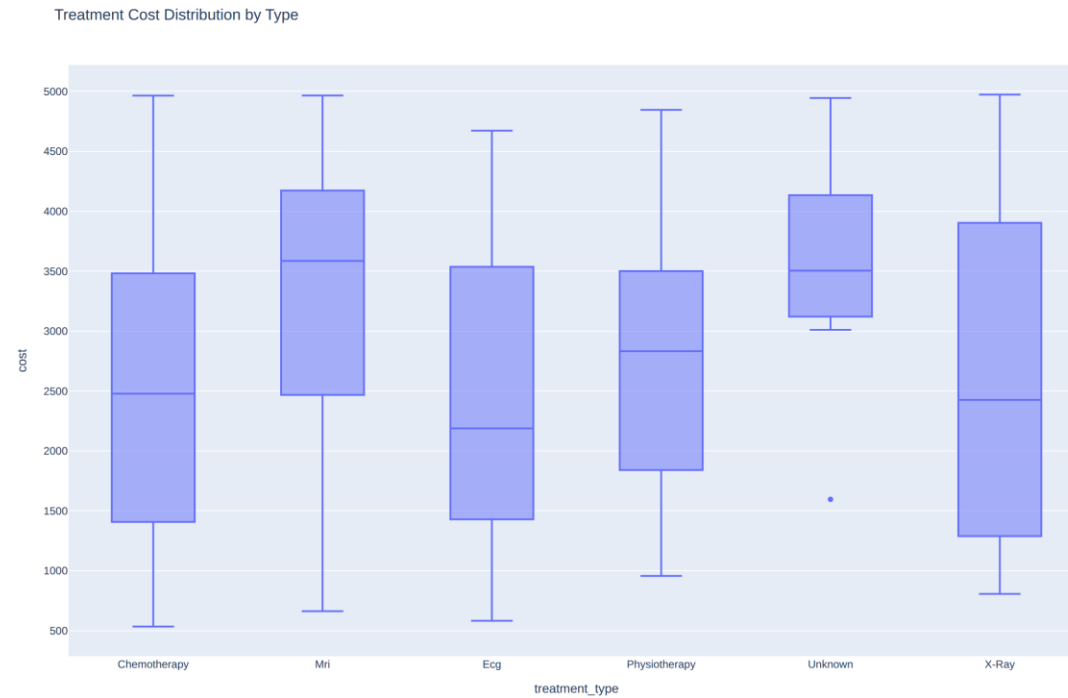
Specialization to Doctor to Treatment Type

Specialization -> Doctor -> Treatment Type



Inner ring: specialization. Middle ring: doctor. Outer ring: treatment type.

Treatment Cost Distribution by Type

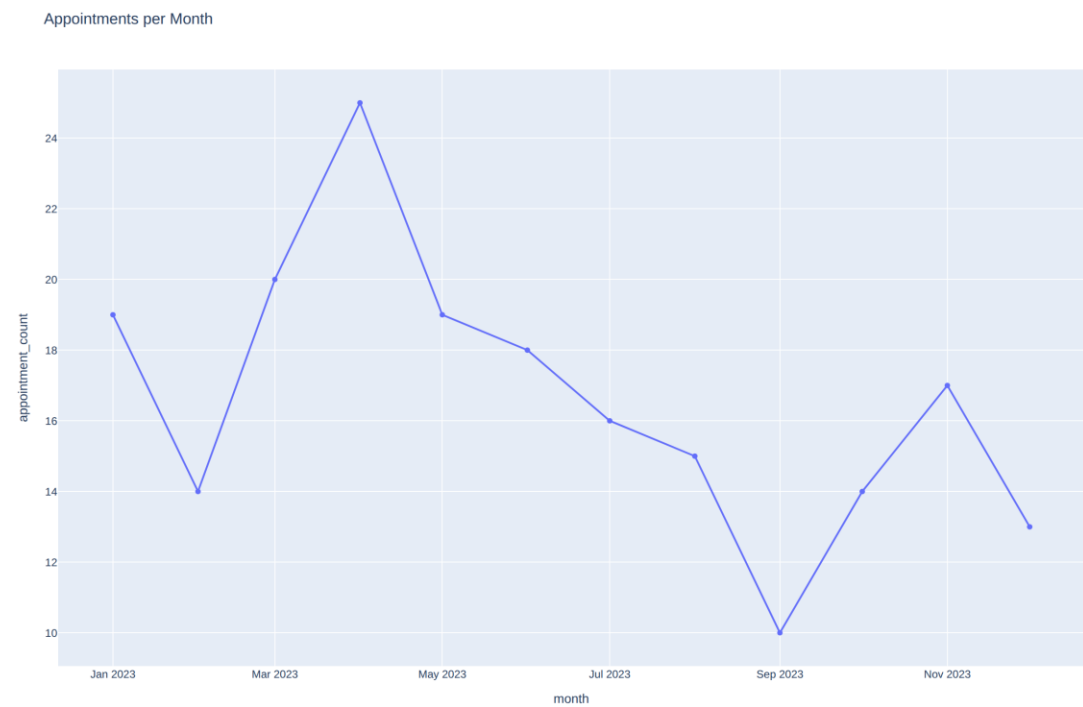


Box plot of treatment cost by treatment type

Cost Distribution: Key Takeaways

- MRI is both the costliest and widest ranging treatment (median Rs. 3,585)
- Unknown treatment type clusters near MRI cost, likely a data entry gap on costly procedures
- Chemotherapy and ECG are the most predictably priced treatments

Monthly Appointment Trend

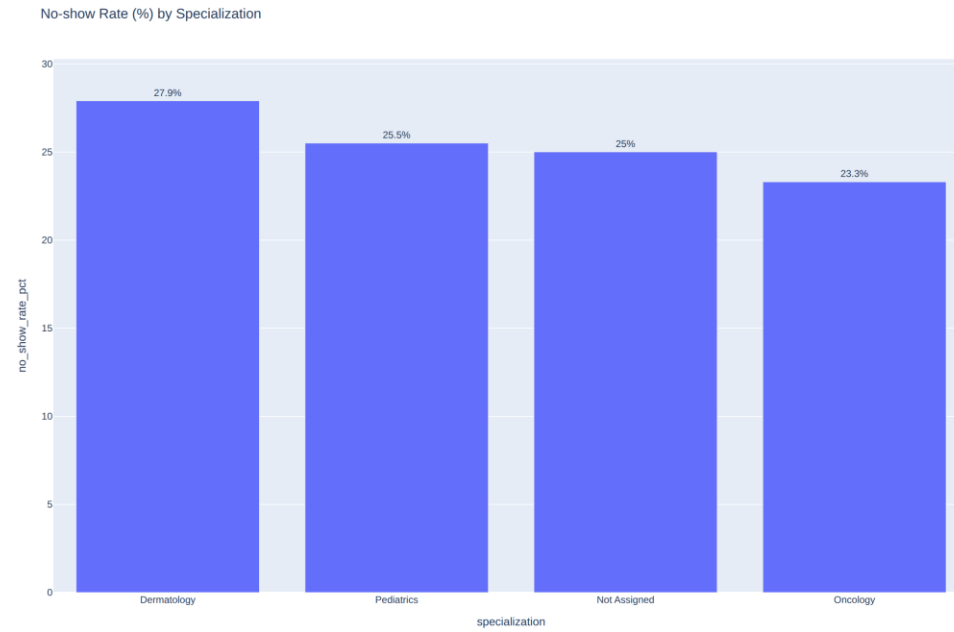


Appointment volume by month across 2023

Monthly Trend: Key Takeaways

- April is the peak month with 25 appointments
- September is the quietest month with just 10 appointments
- A gradual cooling trend follows the April peak through the second half of the year

No-show Rate by Specialization



Percentage of appointments ending in no-show, by specialization

Revenue by Branch

Branch	Appointments	Revenue (Rs.)	Observation
Central Hospital	80	1,95,862.74	Highest revenue and volume
Westside Clinic	52	1,43,244.54	Fewer appointments, costlier mix
Eastside Clinic	60	1,25,222.63	Highest count of smaller branches, lowest revenue

Payment & Billing Analysis

Payment Status	Count	Total Billed (Rs.)	Avg Treatment Cost (Rs.)
Pending	66	1,67,147.63	2,652.39
Failed	61	1,69,780.24	2,863.76
Paid	60	1,48,296.44	2,703.63

Real-World Problems & Solutions

- **Problem: No-shows (26%) outnumber completed visits (23%)**
- Solution: Automated reminders 24 to 48 hours in advance, starting with Dermatology
- **Problem: Rs. 1.69 lakh in failed payments concentrated in costlier treatments**
- Solution: Offer installment options for treatments above the median cost
- **Problem: Dermatology doctors handle the highest per-doctor caseload**
- Solution: Redistribute caseload and pair newer doctors with senior colleagues

Recommendations

- Prioritize a no-show reduction program, starting with Dermatology
- Introduce flexible billing for high-cost treatments
- Track appointments-per-doctor as a standing operational metric
- Close the data entry gap at booking with mandatory doctor assignment
- Investigate the September dip against staff leave and seasonal factors

Conclusion

- 200 appointment records across 5 linked tables turned into actionable operational intelligence
- Biggest opportunities: reducing no-shows, tightening billing collection, matching staffing to demand
- This is a small, deliberately messy sample dataset built to demonstrate a realistic workflow
- Findings should be read as directional patterns, not statistically definitive conclusions